

Rotational Motion

Spinning, rolling and balancing: torque, moment of inertia and angular momentum, the rotational twins of force, mass and momentum.

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Wheels, turbines, hard drives, joints, wrenches, planets and figure skaters all rotate. A **rigid body** keeps its shape, so every point turns through the same angle in the same time. Its motion is completely described by the motion of its centre of mass plus a rotation about it. This chapter develops rotational dynamics by close analogy with linear dynamics: angle for position, torque for force, moment of inertia for mass, angular momentum for momentum. It then applies them to pulleys, rolling bodies, spinning skaters, and the static equilibrium of beams, ladders and the human skeleton.

YOU WILL BE ABLE TO

- Use angular displacement, velocity and acceleration, and relate them to the linear motion of points on the body.
- Calculate torques as $\vec{\tau} = \vec{r} \times \vec{F}$ and use lever arms.
- Compute moments of inertia by integration, and with the parallel- and perpendicular-axis theorems.
- Apply $\sum \tau = I\alpha$ to pulleys with mass, wound strings and falling rods.
- Use rotational kinetic energy, work and power, including for rolling without slipping.
- Apply conservation of angular momentum.
- Solve static equilibrium problems, including biomechanical models of joints and the spine.

THINK FIRST

A solid ball and a hollow ball (a thin spherical shell) have the same mass and radius. They are released together at the top of a ramp and roll down without slipping. Which reaches the bottom first? Does it matter if one ball is heavier than the other?

The **solid ball wins**, and mass and radius do not matter at all. As a ball rolls, some of the gravitational energy goes into spinning it rather than moving it forward. The hollow ball has its mass farther from the axis, so it has a larger moment of inertia ($\frac{2}{3}MR^2$ against $\frac{2}{5}MR^2$). It therefore puts a larger share of its energy into rotation and has less left for forward motion. Only the *shape* (how the mass is distributed) matters, as Example 7.4 shows.

§7.1

Rotational kinematics

Describe the orientation of a body rotating about a fixed axis by an angle θ , measured in **radians**. An arc of length s on a circle of radius r subtends $\theta = s/r$. One revolution is 2π rad, so $1 \text{ rev} = 360^\circ = 2\pi \text{ rad}$.

$$\omega = \frac{d\theta}{dt} \text{ (rad/s)}, \quad \alpha = \frac{d\omega}{dt} \text{ (rad/s}^2\text{)}.$$

Every point of a rigid body has the same θ , ω and α . A point at distance r from the axis has

$$v = r\omega, \quad a_t = r\alpha, \quad a_c = \frac{v^2}{r} = \omega^2 r.$$

Angular velocity is a vector along the axis, with direction given by the right-hand rule: curl your fingers in the direction of rotation and your thumb points along $\vec{\omega}$.

For **constant** α , the equations have exactly the same form as their linear counterparts:

$$\omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2, \quad \omega^2 = \omega_0^2 + 2\alpha\theta.$$

WORKED EXAMPLE 7.1

Spinning up a grinder

An angle grinder's disc goes from rest to 3000 rpm in 5.0 s with constant angular acceleration. The disc has a radius of 6.0 cm. Find (a) its angular acceleration, (b) the

number of revolutions it makes while speeding up, and (c) the speed and the centripetal acceleration of a point on its rim at full speed.

Solution

Convert the final speed: $\omega = 3000 \times \frac{2\pi}{60} = 314 \text{ rad/s}$.

(a) $\alpha = \omega/t = 314/5.0 = 62.8 \text{ rad/s}^2$.

(b) $\theta = \frac{1}{2}\alpha t^2 = \frac{1}{2}(62.8)(25) = 785 \text{ rad}$, which is $785/2\pi = 125$ revolutions.

(c) Rim speed: $v = r\omega = 0.060(314) = 18.8 \text{ m/s}$. Centripetal acceleration:

$$a_c = \omega^2 r = (314)^2(0.060) = 5.9 \times 10^3 \text{ m/s}^2,$$

about 600 g. That is why grinder discs carry a maximum-rpm rating: above it, the disc material cannot supply the centripetal force, and the disc bursts.

§7.2

Torque

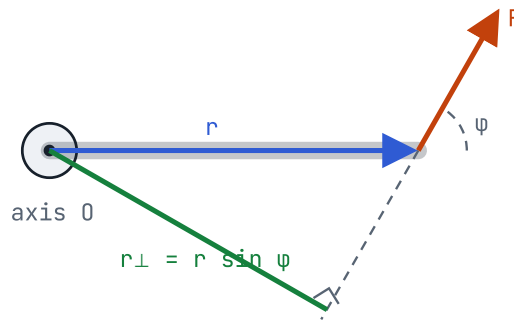
A force's ability to cause rotation depends on its size, its direction, and *where* it is applied. The **torque** about a point O of a force $\vec{F}$ applied at position $\vec{r}$ (measured from O) is

$$\vec{\tau} = \vec{r} \times \vec{F}, \quad \tau = rF \sin \phi = r_{\perp} F = rF_{\perp},$$

where ϕ is the angle between $\vec{r}$ and $\vec{F}$. Torque is measured in N m. (We do not call this unit the joule, because torque is not energy.)

Two equivalent pictures:

- $r_{\perp} = r \sin \phi$ is the **lever arm** (moment arm): the perpendicular distance from the axis to the force's *line of action*.
- $F_{\perp} = F \sin \phi$ is the component of the force perpendicular to $\vec{r}$. Only this component turns the body.



Torque = force $\times$ lever arm. The lever arm $r_{\perp}$ is the perpendicular distance from the axis to the force's line of action.

WORKED EXAMPLE 7.2

Loosening a bolt

A mechanic pulls with 50 N on the end of a wrench 0.25 m long, at 60° to the handle. (a) Find the torque on the bolt. (b) What force, applied perpendicular to the handle, would give the same torque? (c) The bolt needs 40 N m to loosen. Will a 50 N pull do it at any angle?

Solution

(a) $\tau = rF \sin \phi = 0.25(50) \sin 60^\circ = 10.8 \text{ N m}.$

(b) $F_{\perp} = \tau/r = 10.8/0.25 = 43 \text{ N}.$

(c) The largest possible torque, with the pull perpendicular, is $0.25(50) = 12.5 \text{ N m}$, well short of 40 N m. The mechanic needs a longer lever arm: a “cheater bar” of at least $40/50 = 0.80 \text{ m}.$

§7.3

Newton's second law for rotation and the moment of inertia

Consider a rigid body rotating about a fixed axis. A small piece of mass m_i at distance r_i from the axis has tangential acceleration $a_{t,i} = r_i \alpha$. The tangential force on it is $F_i = m_i r_i \alpha$, which gives a

torque $\tau_i = m_i r_i^2 \alpha$. Add these up over all the pieces. The internal torques cancel in pairs, leaving

$$\sum \tau_{\text{ext}} = \left(\sum m_i r_i^2 \right) \alpha = I \alpha.$$

KEY IDEA

Rotational second law: $\sum \tau_{\text{ext}} = I \alpha$, about a fixed axis or about the centre of mass.

Moment of inertia: $I = \sum m_i r_i^2 = \int r^2 dm$, measured in kg m^2 . It plays the role of mass in rotation, but it depends on how the mass is distributed about the chosen axis: mass far from the axis counts for much more.

Calculating moments of inertia

Thin uniform rod (mass M , length L) about a perpendicular axis through one end. With $dm = (M/L) dx$:

$$I_{\text{end}} = \int_0^L x^2 \frac{M}{L} dx = \frac{ML^2}{3}.$$

About its centre, integrate from $-L/2$ to $L/2$ to get $I_{\text{cm}} = \frac{ML^2}{12}$.

Uniform disc (mass M , radius R) about its axis. Split the disc into thin rings of radius r and width dr . Each ring has mass $dm = \frac{M}{\pi R^2} 2\pi r dr$, and all of it lies at distance r from the axis:

$$I = \int_0^R r^2 \frac{2Mr}{R^2} dr = \frac{2M}{R^2} \cdot \frac{R^4}{4} = \frac{1}{2} MR^2.$$

BODY (UNIFORM), AXIS	I
Thin ring or hollow cylinder, central axis	MR^2
Solid disc or cylinder, central axis	$\frac{1}{2} MR^2$
Thick-walled cylinder, radii R_1 and R_2	$\frac{1}{2} M(R_1^2 + R_2^2)$
Solid sphere, through centre	$\frac{2}{5} MR^2$
Thin spherical shell, through centre	$\frac{2}{3} MR^2$

BODY (UNIFORM), AXIS	I
Thin rod, perpendicular axis through centre	$\frac{1}{12} ML^2$
Thin rod, perpendicular axis through end	$\frac{1}{3} ML^2$
Rectangular plate $a \times b$, perpendicular axis through centre	$\frac{1}{12} M(a^2 + b^2)$

KEY IDEA

Parallel-axis theorem. If I_{cm} is the moment of inertia about an axis through the centre of mass, then about any parallel axis a distance d away,

$$I = I_{\text{cm}} + Md^2.$$

Example: a rod about its end has $I = \frac{1}{12} ML^2 + M(L/2)^2 = \frac{1}{3} ML^2$. ✓

Perpendicular-axis theorem (flat plates only). For a lamina in the xy -plane, $I_z = I_x + I_y$.

Example: a thin ring about a diameter has $I = \frac{1}{2} MR^2$, half of its moment about the central axis.

The **radius of gyration** k is defined by $I = Mk^2$. It is the distance from the axis at which all the mass could be concentrated to give the same moment of inertia. Engineers use it to describe the shapes of beams and flywheels.

Pulleys with mass**WORKED EXAMPLE 7.3****An Atwood machine with a heavy pulley**

Repeat the Atwood machine of Example 3.3 ($m_1 = 5.0$ kg, $m_2 = 3.0$ kg), but now the pulley is a uniform disc of mass $M = 2.0$ kg and radius 10 cm. The string does not slip on the pulley. Find the acceleration and both tensions.

Solution

Key change. The tensions on the two sides are now different. Their difference provides the torque that spins the pulley up.

Equations:

$$m_1 : m_1 g - T_1 = m_1 a, \quad m_2 : T_2 - m_2 g = m_2 a, \quad \text{pulley: } (T_1 - T_2)R = I\alpha = \frac{1}{2} M a$$

The rolling condition $a = R\alpha$ links the string to the pulley. The pulley equation simplifies to $T_1 - T_2 = \frac{1}{2} M a$. Adding all three equations:

$$a = \frac{(m_1 - m_2)g}{m_1 + m_2 + \frac{1}{2}M} = \frac{2.0(9.8)}{5.0 + 3.0 + 1.0} = 2.18 \text{ m/s}^2.$$

$$T_1 = m_1(g - a) = 5.0(7.62) = 38.1 \text{ N}, \quad T_2 = m_2(g + a) = 3.0(11.98) = 35.9 \text{ N}.$$

Check: $T_1 - T_2 = 2.2 \text{ N}$, and $\frac{1}{2} M a = 2.18 \text{ N}$. ✓

Evaluate. The pulley's mass enters as an extra “effective mass” of $I/R^2 = \frac{1}{2} M$. The acceleration drops from 2.45 to 2.18 m/s². The massless-pulley result is recovered when $M \rightarrow 0$.

§7.4

Rotational kinetic energy, work and power

Each piece of a rotating body has kinetic energy $\frac{1}{2} m_i (r_i \omega)^2$. Summing:

$$K_{\text{rot}} = \frac{1}{2} I \omega^2.$$

A torque acting through an angle does work, and does it at a rate given by the power:

$$W = \int \tau d\theta, \quad P = \tau \omega.$$

The rotational work-energy theorem is $W_{\text{net}} = \Delta(\frac{1}{2} I \omega^2)$.

LINEAR	ROTATIONAL
x, v, a	θ, ω, α
m	$I = \int r^2 dm$
$F = ma$	$\tau = I\alpha$
$K = \frac{1}{2} m v^2$	$K = \frac{1}{2} I \omega^2$

LINEAR	ROTATIONAL
$W = \int F dx, P = Fv$	$W = \int \tau d\theta, P = \tau\omega$
$\vec{p} = m\vec{v}$	$\vec{L} = I\vec{\omega}$ (about a symmetry axis)
$\vec{F} = d\vec{p}/dt$	$\vec{\tau} = d\vec{L}/dt$

IN PRACTICE

Flywheels, engines and gyroscopes. A flywheel stores energy as $\frac{1}{2}I\omega^2$. Engineers make I large for a given mass by putting the mass in the rim, and make ω as large as the material's strength allows, since the energy grows as ω^2 . Carbon-fibre flywheels spinning at tens of thousands of rpm in vacuum are used for grid frequency regulation, for uninterruptible power supplies in hospitals and data centres, and for energy recovery in racing cars. In an engine, $P = \tau\omega$ links the torque curve to the power curve: an engine making 300 N m at 4000 rpm (419 rad/s) delivers 126 kW. Spinning gyroscopes resist changes in the direction of $\vec{L}$, which is the principle behind gyrocompasses, spacecraft attitude control, and (in micro-machined form) the gyroscopes in phones and drones.

§7.5

Rolling without slipping

A wheel or ball that **rolls without slipping** has its contact point momentarily at rest on the ground. That gives the rolling condition

$$v_{\text{cm}} = R\omega, \quad a_{\text{cm}} = R\alpha.$$

Rolling can be viewed as translation of the centre of mass plus rotation about it, so the kinetic energy is

$$K = \frac{1}{2}Mv_{\text{cm}}^2 + \frac{1}{2}I_{\text{cm}}\omega^2 = \frac{1}{2}Mv_{\text{cm}}^2 \left(1 + \frac{I_{\text{cm}}}{MR^2}\right).$$

Equivalently, the motion is pure rotation about the contact point, with $I_P = I_{\text{cm}} + MR^2$.

The friction at the contact point is **static**, and it does **no work** in pure rolling, because the contact point does not move. So mechanical energy is conserved even though friction acts.

WORKED EXAMPLE 7.4

The rolling race

Bodies of mass M and radius R roll without slipping down an incline of angle θ and height h . Write $I_{\text{cm}} = \beta MR^2$. Find (a) the speed at the bottom, (b) the acceleration, and (c) the friction force needed. Then rank a solid sphere, a solid cylinder, a hollow sphere and a ring.

Solution

(a) Energy.

$$Mgh = \frac{1}{2}Mv^2(1 + \beta) \Rightarrow v = \sqrt{\frac{2gh}{1 + \beta}}.$$

(b) Forces and torques. Along the slope: $Mg \sin \theta - f = Ma$. Torque about the CM: $fR = I\alpha = \beta MR^2(a/R)$, so $f = \beta Ma$. Substituting:

$$a = \frac{g \sin \theta}{1 + \beta}.$$

(c) $f = \beta Ma = \frac{\beta}{1 + \beta} Mg \sin \theta$. Rolling without slipping needs $f \leq \mu_s Mg \cos \theta$, that is,

$$\tan \theta \leq \frac{1 + \beta}{\beta} \mu_s.$$

Ranking (larger a wins):

BODY	β	$a/(g \sin \theta)$
Solid sphere	2/5	5/7 = 0.714
Solid cylinder	1/2	2/3 = 0.667
Hollow sphere	2/3	3/5 = 0.600
Ring or hoop	1	1/2 = 0.500

Evaluate. Mass and radius cancel. Only the shape factor β matters. Any rolling body is slower than a frictionless sliding block ($\beta = 0$), because part of its energy is locked up in spin.

§7.6

Angular momentum

For a particle with momentum $\vec{p}$ at position $\vec{r}$ relative to a point O , the **angular momentum** about O is

$$\vec{L} = \vec{r} \times \vec{p}.$$

For a rigid body rotating about a fixed axis, the component along the axis is $L = I\omega$.

Differentiating $\vec{L} = \vec{r} \times \vec{p}$ gives the general rotational law:

$$\sum \vec{\tau}_{\text{ext}} = \frac{d\vec{L}}{dt}.$$

KEY IDEA

Conservation of angular momentum. If the net external torque on a system is zero, its total angular momentum is constant. For a body whose moment of inertia can change:

$$I_i \omega_i = I_f \omega_f.$$

WORKED EXAMPLE 7.5

A spinning skater

A figure skater spins at 2.0 rev/s with her arms out, with $I = 3.0 \text{ kg m}^2$. She pulls her arms in, reducing I to 1.0 kg m^2 . Find her new spin rate and the change in her kinetic energy.

Solution

The ice exerts almost no torque about the vertical axis, so L is conserved:

$$\omega_f = \frac{I_i}{I_f} \omega_i = 3.0 \times 2.0 = 6.0 \text{ rev/s}.$$

Kinetic energy. Since $K = L^2/2I$ and L is fixed, K is inversely proportional to I . So $K_f = 3K_i$. Numerically, with $\omega_i = 4\pi \text{ rad/s}$:

$$K_i = \frac{1}{2}(3.0)(4\pi)^2 = 237 \text{ J}, \quad K_f = 711 \text{ J}.$$

Where does the extra 474 J come from? From her muscles. Pulling her arms inward against the centrifugal tendency (in her rotating frame) requires work. Angular momentum is conserved, but kinetic energy is not.

WORKED EXAMPLE 7.6**The birth of a pulsar**

The core of a massive star, with a radius of about 7×10^5 km (like the Sun) and a rotation period of 25 days, collapses to form a neutron star of radius 10 km. Treat both as uniform spheres of the same mass. Estimate the neutron star's rotation period.

Solution

$L = \frac{2}{5}MR^2\omega$ is conserved, so $\omega \propto 1/R^2$ and $T \propto R^2$:

$$T_f = T_i \left(\frac{R_f}{R_i} \right)^2 = (25 \times 86\,400 \text{ s}) \left(\frac{10}{7 \times 10^5} \right)^2 = 2.16 \times 10^6 \times 2.0 \times 10^{-10} = 4.4 \times 10^{-4} \text{ s}$$

Evaluate. The star spins about 2000 times a second, which is in the right range: observed pulsars rotate with periods from about 1.4 ms to several seconds. Real collapses shed angular momentum, so typical new pulsars are slower. This was a key clue in identifying pulsars, discovered in 1967 by Jocelyn Bell Burnell, as spinning neutron stars.

§7.7

Static equilibrium

A rigid body is in **static equilibrium** when it is at rest and stays at rest. This requires both

$$\sum \vec{F} = 0 \quad \text{and} \quad \sum \vec{\tau} = 0 \text{ about any point.}$$

Because the net torque must vanish about *every* point, you may choose the point that makes the algebra easiest. Choose the point where unknown forces act, so that they drop out of the torque equation.

KEY IDEA

Strategy for statics.

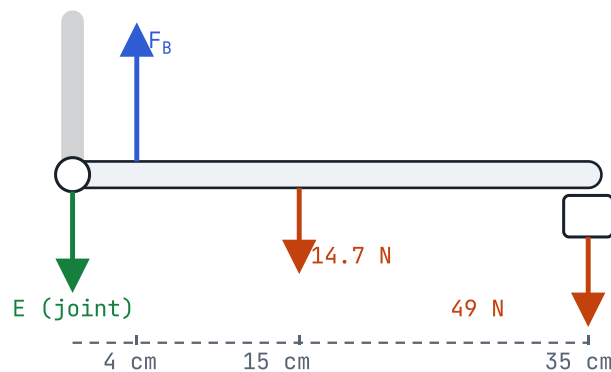
1. Draw the FBD of the body, showing *where* each force acts.
2. Choose a pivot point where one or more unknown forces act.
3. Write $\sum \tau = 0$ about that point, taking counter-clockwise as positive.
4. Write $\sum F_x = 0$ and $\sum F_y = 0$.
5. Solve, and check with a torque equation about a different point.

WORKED EXAMPLE 7.7

The biceps and the elbow

A person holds a 5.0 kg dumbbell in the hand with the forearm horizontal. The forearm and hand have a mass of 1.5 kg, with their centre of mass 15 cm from the elbow joint. The dumbbell is 35 cm from the elbow. The biceps tendon attaches to the forearm 4.0 cm from the elbow and pulls vertically upward. Find the force in the biceps and the force at the elbow joint.

Solution



Forearm as a lever. The biceps acts very close to the pivot at the elbow, so it must pull far harder than the load it holds.

Torques about the elbow. This eliminates the unknown joint force E . Counter-clockwise is positive:

$$F_B(0.040) - (1.5)(9.8)(0.15) - (5.0)(9.8)(0.35) = 0.$$

$$F_B = \frac{2.21 + 17.15}{0.040} = \frac{19.36}{0.040} = 484 \text{ N}.$$

Vertical forces, taking E as positive upward:

$$F_B - 14.7 - 49 + E = 0 \Rightarrow E = -420 \text{ N}.$$

The joint force is 420 N, pushing **down** on the forearm.

Evaluate. The biceps pulls with about 7.6 times the combined weight of the dumbbell and forearm, because its lever arm (4 cm) is so much shorter than the load's (35 cm). The body trades force for speed and range of motion: a small contraction of the biceps moves the hand a long way, quickly. The price is very large muscle and joint forces. This is why tendon and joint injuries are so common, and why orthopaedic surgeons and the designers of prosthetic joints must plan for loads many times body weight.

WORKED EXAMPLE 7.8

Will the ladder slip?

A uniform 5.0 m ladder of mass 20 kg leans against a frictionless wall at 60° to the floor. Find the minimum coefficient of static friction at the floor for the ladder not to slip.

Solution

Forces: the weight mg at the midpoint; the floor's normal force N_f and friction f at the base; the wall's normal force N_w (horizontal) at the top.

Force balance: $N_f = mg$ and $f = N_w$.

Torques about the base (eliminating N_f and f). With the ladder at angle θ to the floor, the lever arm of the weight is $(L/2) \cos \theta$ and that of N_w is $L \sin \theta$:

$$N_w L \sin \theta = mg \frac{L}{2} \cos \theta \Rightarrow N_w = \frac{mg}{2 \tan \theta}.$$

Friction needed:

$$f = N_w \leq \mu_s N_f = \mu_s mg \Rightarrow \mu_s \geq \frac{1}{2 \tan \theta} = \frac{1}{2 \tan 60^\circ} = 0.29.$$

Evaluate. A shallower ladder (smaller θ) needs more friction. Safety guidance recommends a 4:1 ratio of height to base distance, about 75° , for which $\mu_s \geq 0.13$. A

person climbing the ladder raises the required friction as they go higher (Problem P7.11), which is why ladders tend to slip when the climber is near the top.

IN PRACTICE

Biomechanics: why “lift with your legs”. When you bend forward to lift a load, the spine acts as a lever pivoted at the lumbosacral joint (L5/S1). The back muscles (erector spinae) pull nearly parallel to the spine, with a lever arm of only a few centimetres, while the weight of the upper body and the load act with lever arms of tens of centimetres. The muscle force, and the resulting compression of the intervertebral disc, can reach several thousand newtons (Problem P7.17). Ergonomic guidelines limit disc compression to about 3.4 kN. Keeping the back upright and the load close to the body cuts the load’s lever arm, and with it the spinal force. This is the physics behind workplace manual-handling rules, and an everyday example of static equilibrium.

CHAPTER SUMMARY

- $\omega = d\theta/dt$ and $\alpha = d\omega/dt$. A point at radius r has $v = r\omega$, $a_t = r\alpha$ and $a_c = \omega^2 r$. The constant- α equations mirror the linear ones.
- Torque: $\vec{\tau} = \vec{r} \times \vec{F}$, with magnitude $rF \sin \phi = r_{\perp} F$.
- $\sum \tau = I\alpha$, with $I = \int r^2 dm$. Parallel axis: $I = I_{\text{cm}} + Md^2$. Perpendicular axis (laminae): $I_z = I_x + I_y$.
- $K_{\text{rot}} = \frac{1}{2} I\omega^2$ and $P = \tau\omega$.
- Rolling without slipping: $v = R\omega$, $K = \frac{1}{2} Mv^2(1 + \beta)$ and $a = g \sin \theta / (1 + \beta)$, where $\beta = I/MR^2$.
- $\vec{L} = \vec{r} \times \vec{p}$, or $L = I\omega$ for a rigid body. $\vec{\tau} = d\vec{L}/dt$. With no external torque, L is conserved.
- Static equilibrium: $\sum \vec{F} = 0$ and $\sum \vec{\tau} = 0$ about any point. Take torques about the point where the unknown forces act.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P7.1

(a) Can a body be rotating if no torque acts on it? (b) Can a body have zero net force on it but a non-zero net torque? Give an example of each, or explain why not.

P7.2

A solid cylinder and a hollow cylinder (a thin-walled pipe) of equal mass and radius roll without slipping down the same ramp. Which reaches the bottom first? Would the answer change if the hollow one were twice as heavy?

P7.3

Explain why a figure skater spins faster when she pulls her arms in. Is her kinetic energy conserved? If not, where does the change come from?

P7.4

Why are door handles placed on the side opposite the hinges? Estimate how much harder it is to open a door by pushing 10 cm from the hinges instead of at the handle, 80 cm from them.

LEVEL B – STANDARD PROBLEMS

P7.5

A flywheel is a uniform disc of mass 50 kg and radius 0.40 m spinning at 3000 rpm. (a) Find its kinetic energy. (b) What constant braking torque would stop it in 20 s? (c) How many revolutions does it make while stopping?

P7.6

Four small 0.50 kg masses sit at the corners of a square of side 0.40 m, joined by light rods. Find the moment of inertia about (a) an axis through the centre of the square, perpendicular to its plane, (b) an axis along one side of the square, and (c) an axis along a diagonal.

P7.7

A 5.0 kg bucket hangs from a light rope wound around a windlass: a solid cylinder of mass 10 kg and radius 0.10 m, free to turn on a frictionless axle. The bucket is released from rest. Find (a) its acceleration, (b) the tension in the rope, and (c) its speed after falling 3.0 m.

P7.8

A solid ball rolls without slipping, starting from rest at the top of a ramp 1.5 m high. Find its speed at the bottom. Compare this with the speed of a block sliding down a frictionless ramp of the same height, and explain the difference.

P7.9

A 30 kg child sits 1.5 m from the pivot of a seesaw. Where must a 40 kg child sit to balance it? What force does the pivot exert if the plank has a mass of 10 kg and is balanced at its centre?

P7.10

Holding the arm out. A person holds their 4.0 kg arm straight out horizontally. The arm's centre of mass is 0.30 m from the shoulder joint. The deltoid muscle attaches 0.15 m from the joint and pulls at 15° above the arm. Find the tension in the deltoid and the magnitude of the force at the shoulder joint.

P7.11

A painter of mass 70 kg climbs a uniform 5.0 m ladder of mass 20 kg, which leans at 60° to the floor against a frictionless wall. The coefficient of static friction at the floor is 0.40. How far up the ladder can the painter climb before it slips?

P7.12

A playground merry-go-round is a uniform disc of mass 200 kg and radius 2.0 m, turning freely once every 4.0 s. A 50 kg child starts at the rim and walks to the centre. (a) Find the final rotation period. (b) Find the change in kinetic energy and explain where the energy comes from.

LEVEL C – CHALLENGE PROBLEMS**P7.13**

Derive the moment of inertia of a uniform solid sphere about a diameter, $I = \frac{2}{5}MR^2$, by slicing it into thin discs perpendicular to the axis.

P7.14

A bowling ball of radius R is launched along the lane with speed v_0 and no spin. The coefficient of kinetic friction between ball and lane is μ_k . (a) Show that the ball slides for a time $t = 2v_0/(7\mu_k g)$ before it starts to roll without slipping. (b) Show that its rolling speed is then $\frac{5}{7}v_0$. (c) Find the distance it slides, and evaluate everything for $v_0 = 8.0$ m/s and $\mu_k = 0.20$. (d) What fraction of the initial kinetic energy is lost?

P7.15

A uniform rod of length L stands vertically on the floor. Its lower end is held in place by a hinge, and it is allowed to fall over. (a) Find its angular speed when it hits the floor. (b) Show that the free end then moves faster than an object dropped from height L . (c) Show that, when the rod is horizontal, the free end has an acceleration of $\frac{3}{2}g$, faster than free fall. Explain why tall falling chimneys often break in two partway up.

P7.16

A yo-yo is a uniform disc of mass M and radius R . A string is wound around it, and the free end is held fixed. The yo-yo is released and unwinds as it falls. Find its acceleration and the tension in the string.

P7.17

Lifting with a bent back. A 75 kg person bends forward so that the back is at 30° above the horizontal. Model the spine as a rigid rod of length L pivoted at the base (L5/S1). The weight of the upper body (450 N) acts at the midpoint of the spine. A 200 N load held in the hands acts at the top of the spine. The erector spinae muscles attach two-thirds of the way up the spine and pull at 12° to it. Find (a) the muscle tension and (b) the compressive force along the spine at L5/S1. Compare the compressive force with the person's body weight and with the 3.4 kN guideline limit.

P7.18

A uniform rod of mass 2.0 kg and length 1.0 m hangs vertically from a frictionless pivot at its top end. A 10 g bullet travelling horizontally at 300 m/s hits the bottom of the rod and embeds itself. (a) Which quantity is conserved during the collision: linear momentum, angular momentum about the pivot, or kinetic energy? Explain. (b) Find the angular velocity just after the impact. (c) Find the maximum angle through which the rod swings.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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